## SOMMA GEOMETRICA

$$\sum_{k=0}^{n} q^k = q^0 + q^1 + \dots + q^n = \frac{1-q^{n+1}}{1-q} \qquad (\#)$$

$q \neq 1, 0$

$$\sum_{k=0}^{n} 1 = n+1$$

### DIM PER INDUZIONE

BASE DELL'INDUZIONE CIOÈ $n=0$

$$1 = q^0 = \sum_{k=0}^{0} q^k \stackrel{?}{=} \frac{1-q^{0+1}}{1-q} = \frac{1-q}{1-q} = 1$$

IPOTESI INDUTTIVA CIOÈ LA FORMULA (#) LA SUPPONIAMO VERA DA 0 A $n$

DIMOSTRARE LA FORMULA PER $n+1$

$$\sum_{k=0}^{n+1} q^k = (1+q+\dots+q^n)+q^{n+1} = \left(\sum_{k=0}^{n} q^k\right) + q^{n+1}$$

$$= \left(\frac{1-q^{n+1}}{1-q}\right) + q^{n+1}$$

$$= \frac{1-q^{n+1}+(q^{n+1})(1-q)}{1-q}$$

$$= \frac{1-q^{n+1}+q^{n+1}-q^{n+2}}{1-q}$$

$$= \frac{1-q^{n+2}}{1-q} = \frac{1-q^{(n+1)+1}}{1-q}$$

## ESERCIZIO PRE-ESONERO

$$a_n = \frac{n^5+A}{n^5+8}, \quad A \in \mathbb{R}$$

$$\frac{n^5+A}{n^5+8} = \frac{n^5+8-8+A}{n^5+8} = 1 + \frac{A-8}{n^5+8}$$

$$\downarrow$$

$$b_n = \frac{A-8}{n^5+8} \qquad A=8 \Rightarrow a_n = 1$$

$A > 8$ $a_n \searrow$

$A = 8$ $a_n$ COSTANTE

$A > 8$ $a_n \nearrow$

$A > 8$

$b_n = \frac{A-8}{n^5+8}$

$b_{n+1} \frac{A-8}{(n+1)^5+8}$

$b_n > b_{n+1} \iff \frac{A-8}{n^5+8} > \frac{A-8}{(n+1)^5+8}$

$\frac{1}{n^5+8} > \frac{1}{(n+1)^5+8}$

$(n+1)^5+8 > n^5+8$

$(n+1)^5 > n^5$ Vera

$b_n \searrow$ quindi anche $a_n$ che è

$a_n = 1 + b_n$

$a_n \searrow$

$A < 8$

$b_n < b_{n+1}$ $\frac{A-8}{n^5+8} < \frac{A-8}{(n+1)^5+8}$

$(n+1)^5+8 > n^5+8$

$(n+1)^5 > n^5+8$

$a_n \nearrow$

1) $\lim_{n\to\infty} a_n = 1$ ? Vera

" $= \frac{n^5+A}{n^5+8}$

$= \frac{n^5\left(1+\frac{A}{n^5}\right)}{n^5\left(1+\frac{8}{n^5}\right)} = 1$

2) Se $A=11$ $a_n \geq 1$ $\forall n \in \mathbb{N}$ ? Vera

$a_n \searrow$

$\lim_{n\to\infty} a_n = 1 = \inf(a_n) \Rightarrow a_n \geq 1 \ \forall n \in \mathbb{N}$

3) $A=10$ $a_n \searrow$ ? Vera

4) $A=-8$ $a_n \nearrow$ ? Vera

ESERCIZIO

$\lim_{n\to\infty} \sqrt{n+1} - \sqrt{n} \quad \to +\infty - \infty$

$\lim_{n\to\infty} (\sqrt{n+1} - \sqrt{n}) \left(\frac{\sqrt{n+1} + \sqrt{n}}{\sqrt{n+1} + \sqrt{n}}\right)$

$\lim_{n\to\infty} (n+1-n) \cdot \frac{1}{\sqrt{n+1}+\sqrt{n}}$

$\lim_{n\to\infty} 1 \cdot \frac{1}{\sqrt{n+1}+\sqrt{n}} = \frac{1}{\infty+\infty} = 0$

ESERCIZIO

$E = \{x \in \mathbb{R} : x^2 \leq 7\}$
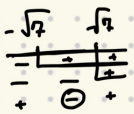
D: $x^2 \leq 7$

$x = -\sqrt{7}, \sqrt{7}$

$1(x+\sqrt{7})(x-\sqrt{7}) \leq 0$

$x+\sqrt{7} \geq 0 \to x \geq -\sqrt{7}$

$x-\sqrt{7} \geq 0 \to x \geq \sqrt{7}$




$-\sqrt{7} \ x \leq \sqrt{7}$

$E = [-\sqrt{7}, \sqrt{7}]$

A. E è un intervallo? Vero

B. E è limitato? Vero

C. $\exists \max(E)$? Vero $\sqrt{7}$

D. $\nexists \min(F = E \cap \mathbb{Q})$? Vero

$-\sqrt{7} \in E$

$-\sqrt{7} \in \mathbb{Q}$ ?

$-\sqrt{7} = p/q$

$7 = \frac{p^2}{q^2}$

$7q^2 = p^2 \quad p = 7k$

$7q^2 = (7k)^2$

$7q^2 = 49k^2 \quad -\sqrt{7} \notin \mathbb{Q}$

$q^2 = 7k^2$

ESERCIZIO

$D\left(\frac{1}{x-2}\right)$

↑ Dominio $dom(f)$, $D(f)$

D: $x \in \mathbb{R} \setminus \{2\}$

$x \neq 2$

$D\left(\sqrt[5]{4-x^2}\right)$

D: $x \in \mathbb{R}$

$D\left(\log_8\left(\frac{2-x}{x+2}\right)\right)$

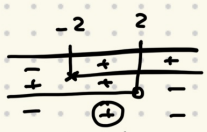
D: $\frac{2-x}{x+2} > 0$

$x+2 \neq 0$

$x \neq -2$

$-x+2 > 0 \to x < 2$

$x+2 > 0 \to x > -2$



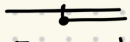

$(-2, 2)$

ESERCIZIO

$D\left(\sqrt[10]{x+5} - \sqrt[5]{x-5}\right)$

$D: \quad x+5 \geq 0$

$x \geq -5$




$[-5, +\infty)$

ESERCIZIO

$\lim_{n\to\infty} 3n \cdot \sin\frac{6}{n} = \frac{\infty}{\infty}$

$\vartheta_n \to 0, \quad \frac{\sin(\vartheta_n)}{\vartheta_n} \to 1$

$\lim_{n\to\infty} 3n \cdot \frac{\sin\frac{6}{n}}{\frac{6}{n}} \cdot \frac{6}{n}$ (il rapporto $\to 1$)

$\lim_{n\to\infty} 3\cancel{n} \cdot 1 \cdot \frac{6}{\cancel{n}} = 18$

ESERCIZIO

$\lim_{n\to\infty} \frac{\sin(5n)}{3n} = \frac{\infty}{\infty}$

$5n \not\to 0$ quindi non possiamo usare il limite notevole dell'esercizio precedente

$\frac{-1}{3n} \leq \frac{\sin(5n)}{3n} \leq \frac{1}{3n}$

$\to 0 \qquad\qquad \to 0$

Teorema dei carabinieri

quindi anche

$\lim_{n\to\infty} \frac{\sin(5n)}{3n} = 0$